

Randomized and low-rank approximation

Clustered singular-value gaps in randomized block Krylov approximation

Difficulty: hard **Importance:** interesting to specialist**Status:** Partially resolved

Rating rationale: Hard because this is a focused spectral-gap refinement of an established Krylov bound; specialist impact reflects the source's assessment that its practical gain is limited.

Partial results — 2026-09-12

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation ([verified affiliation](#)). The [complete report](#) and [editable source](#) establish the following partial results, with $m = k' = bt$ and $\Delta = \Delta_{k'}^{(b)}$.

Theorem 5.3 proves all three requested guarantees for every admissible input with

$$q = O\left(\frac{t \log(2m/\Delta) + \log(n/(\delta\epsilon))}{\sqrt{\epsilon}}\right).$$

This has an extra $t \log m$ term. The requested iteration order holds when $\Delta \leq 1/m$ (Corollary 5.4), when the leading spectrum consists of exactly t levels each repeated b times (Theorem 6.1), when $t = 2$ with arbitrary admissible leading spectra (Theorem 7.1), and at the endpoints $b = 1$ and $b = k$ (Theorem 6.1 and Corollary 6.2). Proposition 8.1 gives almost-sure exact optimal approximation when $\text{rank}(A) = m$ after $t + 1$ blocks. Theorem 4.1 supplies an alternative all-input bound with leading term $(m - b + 1) \log(3/\Delta)$.

Remaining target: the displayed original bound for arbitrary growing b, t , nonzero-width leading clusters and $\Delta > 1/m$ is neither proved nor disproved. The report's sufficient interpolation estimate (IE) remains unproved. Its raw-monomial conditioning counterexample (Proposition 9.1) does not refute RA-04, whose algorithm depends on the Krylov subspace.

A [separate independent Codex agent review](#) passed the partial arguments and checked the imported convergence theorem against the primary source. This is informal AI-agent review, not external human peer review or formal verification. It does not meet the complete-resolution requirement for **Solved**. [Submission and reproduction record](#).

Original problem (retained)

Let $A \in \mathbb{R}^{n \times d}$ have singular values $\sigma_1 \geq \sigma_2 \geq \dots$. Fix integers $1 \leq b \leq k$, set $t = \lceil k/b \rceil$, and set $k' = bt \leq \text{rank}(A)$. Define

$$\Delta_{k'}^{(b)} = \min_{1 \leq i \leq k' - b} \frac{\sigma_i^2 - \sigma_{i+b}^2}{\sigma_i^2},$$

with the empty minimum set to one, and suppose this gap is positive. Draw $G \in \mathbb{R}^{n \times b}$ with independent standard Gaussian entries. In exact arithmetic, let Z be an orthonormal basis of

$$\text{range}[G, (AA^T)G, \dots, (AA^T)^{q-1}G], \quad \hat{A} = Z[Z^T A]_k,$$

where $[B]_k$ denotes a best rank- k approximation obtained by truncating the SVD of B .

Does an absolute constant C exist such that, for every such input and every $0 < \epsilon, \delta < 1/2$, taking

$$q = \left\lceil C \left[\frac{t}{\sqrt{\varepsilon}} \log \frac{2}{\Delta_{k'}^{(b)}} + \frac{1}{\sqrt{\varepsilon}} \log \frac{n}{\delta \varepsilon} \right] \right\rceil$$

gives, with probability at least $1 - \delta$, both

$$\|A - \widehat{A}\|_{\xi} \leq (1 + \varepsilon) \|A - [A]_k\|_{\xi} \quad (\xi = 2, F),$$

and $|||Av_i||_2^2 - \sigma_i^2| \leq \varepsilon \sigma_{k+1}^2$ for the ordered top k right singular vectors v_i of $\widehat{A}$? Here $\sigma_{k+1} = 0$ if necessary. The factor two makes the logarithm meaningful at a unit gap.

This is the gap-independent algorithmic component of the authors' concluding conjecture. They also conjecture the corresponding improvement in their gap-dependent theorem and random Krylov matrix conditioning theorem; those are not separately counted here. The displayed parameter regime is where the source's spectral-gap expression is finite and defined.

The existing theorem instead depends logarithmically on every consecutive gap and a leading spectral condition number. The proposed bound would explain why a block can accommodate clusters of up to b singular values. It does not ask for a finite-precision extension.

References

1. T. Chen, E. N. Epperly, R. A. Meyer, C. Musco, and A. Rao, *Does block size matter in randomized block Krylov low-rank approximation?*, SODA 2026, pp. 1026–1046 ([published paper](#)); arXiv:2508.06486v2. Section 5 states the conjecture; Section 3.3 gives the full quantitative Theorem 1.3, and Algorithm 1 fixes the algorithm. [Paper](#).
2. R. A. Meyer, C. Musco, and C. Musco, *On the Unreasonable Effectiveness of Single Vector Krylov Methods for Low-Rank Approximation*, SODA 2024, pp. 811–845, especially Theorem 4.5. [Paper](#).

Status check — 2026-09-08

Searched the exact 2025 paper title, “block Krylov b-th order gap conjecture”, and “randomized block Krylov clustered gaps 2026”; checked the current arXiv abstract/version record and Section 5. No proof or counterexample was found. Input perturbation results in Section 3.5 do not establish the unperturbed statement above.

Audit — 2026-09-10

Rechecked §5 of the latest arXiv v2, which retains the clustered-gap conjecture, and added the [SODA 2026 publication](#), pp. 1026–1046. Clustered-gap and later low-rank-query searches found no resolution. Randomly perturbing the input changes the displayed target.